# Project 3

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2026-03-13

## Contents

### Introduction

I am researching the relationship between MLB payroll over time as well as payroll relationship to wins. I will be using data collected from https://www.stevetheump.com/Payrolls.htm, as well as https://www.baseball-reference.com. I have 20 years of data from 2006-2025.

MLB payrolls have been increasing over time as players are getting paid more and more with record contracts, the highest payroll in the MLB currently is the Dodgers, whose payroll is more than two times the median MLB payroll. They have also won the last 2 MLB titles. I would like to investigate the trend of the total MLB payroll as well as explore this relationship while considering things like past years statistics.

### Data

```
wins_data=read_excel("../../data/mlb_data.xlsx", sheet="Wins")
payroll_data=read_excel("../../data/mlb_data.xlsx", sheet="Payroll")
OPS_data=read_excel("../../data/mlb_data.xlsx", sheet="OPS")
ERA_data=read_excel("../../data/mlb_data.xlsx", sheet="ERA")
head(wins_data)
```

```
## # A tibble: 6 x 31
##    Year Dodgers Yankees Astros Braves Phillies Mariners Guardians  Rays
##   <dbl>   <dbl>   <dbl>  <dbl>  <dbl>    <dbl>    <dbl>     <dbl> <dbl>
## 1  2025      93      94     87     76       96       90        88    77
## 2  2024      98      94     88     89       95       85        92    80
## 3  2023     100      82     90    104       90       88        76    99
## 4  2022     111      99    106    101       87       90        92    86
## 5  2021     106      92     95     88       82       90        80   100
## 6  2020      43      33     29     35       28       27        35    40
## # i 22 more variables: 'Blue Jays' <dbl>, Padres <dbl>, Mets <dbl>,
## #   Dbacks <dbl>, Cubs <dbl>, 'Red Sox' <dbl>, Tigers <dbl>, Royals <dbl>,
## #   Rangers <dbl>, Twins <dbl>, Giants <dbl>, Orioles <dbl>, Brewers <dbl>,
## #   Reds <dbl>, Cardinals <dbl>, Angels <dbl>, Pirates <dbl>, Nationals <dbl>,
## #   Marlins <dbl>, Athletics <dbl>, 'White Sox' <dbl>, Rockies <dbl>
```

```
head(payroll_data)
```

```
## # A tibble: 6 x 32
##    Year Dodgers Yankees Astros Braves Phillies Mariners Guardians  Rays
##   <dbl>   <dbl>   <dbl>  <dbl>  <dbl>    <dbl>    <dbl>     <dbl> <dbl>
```

```
## 1  2025     325     315    245    230      250      150       105    95
## 2  2024     310     305    240    225      245      145       100    90
## 3  2023     223     278    193    203      243      140        92    78
## 4  2022     261     250    183    181      230      115        68    86
## 5  2021     262     201    187    148      182       90        80   100
## 6  2020     108     109     82     62       78       41        36    28
## # i 23 more variables: `Blue Jays` <dbl>, Padres <dbl>, Mets <dbl>,
## #   Dbacks <dbl>, Cubs <dbl>, `Red Sox` <dbl>, Tigers <dbl>, Royals <dbl>,
## #   Rangers <dbl>, Twins <dbl>, Giants <dbl>, Orioles <dbl>, Brewers <dbl>,
## #   Reds <dbl>, Cardinals <dbl>, Angels <dbl>, Pirates <dbl>, Nationals <dbl>,
## #   Marlins <dbl>, Athletics <dbl>, `White Sox` <dbl>, Rockies <dbl>,
## #   Total <dbl>
```

```
head(OPS_data)
```

```
## # A tibble: 6 x 31
##    Year Dodgers Yankees Astros Braves Phillies Mariners Guardians  Rays
##   <dbl>   <dbl>   <dbl>  <dbl>  <dbl>    <dbl>    <dbl>     <dbl> <dbl>
## 1  2025   0.764   0.783  0.714  0.72     0.756    0.738     0.667 0.714
## 2  2024   0.779   0.761  0.738  0.722    0.746    0.687     0.702 0.669
## 3  2023   0.795   0.701  0.768  0.824    0.762    0.734     0.695 0.776
## 4  2022   0.775   0.751  0.745  0.761    0.739    0.704     0.699 0.69
## 5  2021   0.759   0.719  0.783  0.754    0.726    0.685     0.707 0.749
## 6  2020   0.821   0.789  0.72   0.832    0.781    0.678     0.662 0.753
## # i 22 more variables: `Blue Jays` <dbl>, Padres <dbl>, Mets <dbl>,
## #   Dbacks <dbl>, Cubs <dbl>, `Red Sox` <dbl>, Tigers <dbl>, Royals <dbl>,
## #   Rangers <dbl>, Twins <dbl>, Giants <dbl>, Orioles <dbl>, Brewers <dbl>,
## #   Reds <dbl>, Cardinals <dbl>, Angels <dbl>, Pirates <dbl>, Nationals <dbl>,
## #   Marlins <dbl>, Athletics <dbl>, `White Sox` <dbl>, Rockies <dbl>
```

```
head(ERA_data)
```

```
## # A tibble: 6 x 31
##    Year Dodgers Yankees Astros Braves Phillies Mariners Guardians  Rays
##   <dbl>   <dbl>   <dbl>  <dbl>  <dbl>    <dbl>    <dbl>     <dbl> <dbl>
## 1  2025    5.01    3.63   3.56   4.31     4.97     7.14      4.93  3.08
## 2  2024    3.9     3.74   3.71   3.49     3.81     3.49      3.61  3.78
## 3  2023    4.06    3.97   3.94   4.14     3.97     3.74      3.97  3.86
## 4  2022    2.8     3.3    2.9    3.44     3.97     3.59      3.46  3.41
## 5  2021    3.01    3.74   3.78   3.88     4.39     4.3       4.34  3.67
## 6  2020    3.02    4.35   4.31   4.41     5.14     5.03      3.17  3.56
## # i 22 more variables: `Blue Jays` <dbl>, Padres <dbl>, Mets <dbl>,
## #   Dbacks <dbl>, Cubs <dbl>, `Red Sox` <dbl>, Tigers <dbl>, Royals <dbl>,
## #   Rangers <dbl>, Twins <dbl>, Giants <dbl>, Orioles <dbl>, Brewers <dbl>,
## #   Reds <dbl>, Cardinals <dbl>, Angels <dbl>, Pirates <dbl>, Nationals <dbl>,
## #   Marlins <dbl>, Athletics <dbl>, `White Sox` <dbl>, Rockies <dbl>
```

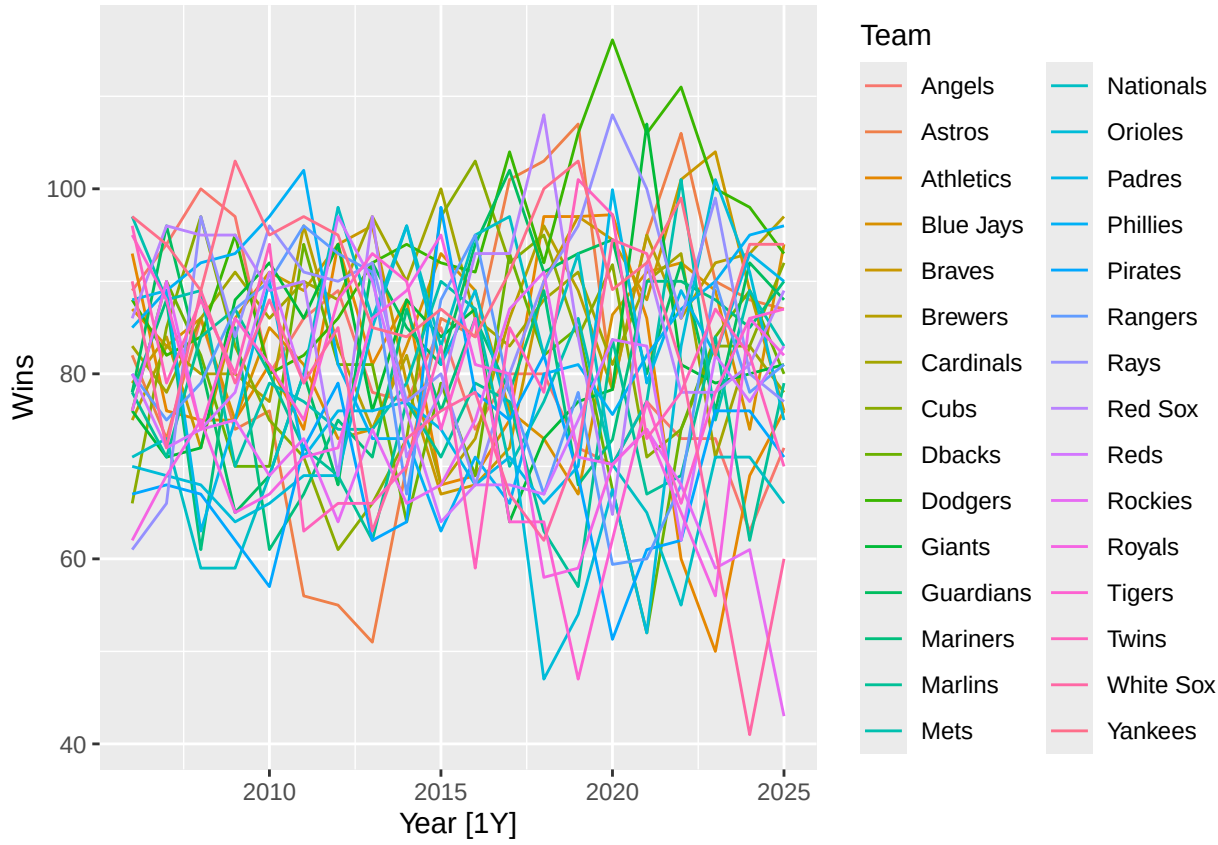
```
# Scale row 6 (2020 season) by 162/60; keep rest unchanged. Then make entire dataset a time series.
wins_ts <- wins_data |>
  pivot_longer(-Year, names_to = "Team", values_to = "Wins") |>
  mutate(Wins = if_else(Year == 2020, Wins * (162 / 60), Wins)) |>
  as_tsibble(key = Team, index = Year)
```

```
payroll_ts <- payroll_data |>
    pivot_longer(-Year, names_to = "Team", values_to = "Payroll") |>
    mutate(Payroll = if_else(Year == 2020, Payroll * (162 / 60), Payroll)) |>
    as_tsibble(key = Team, index = Year)

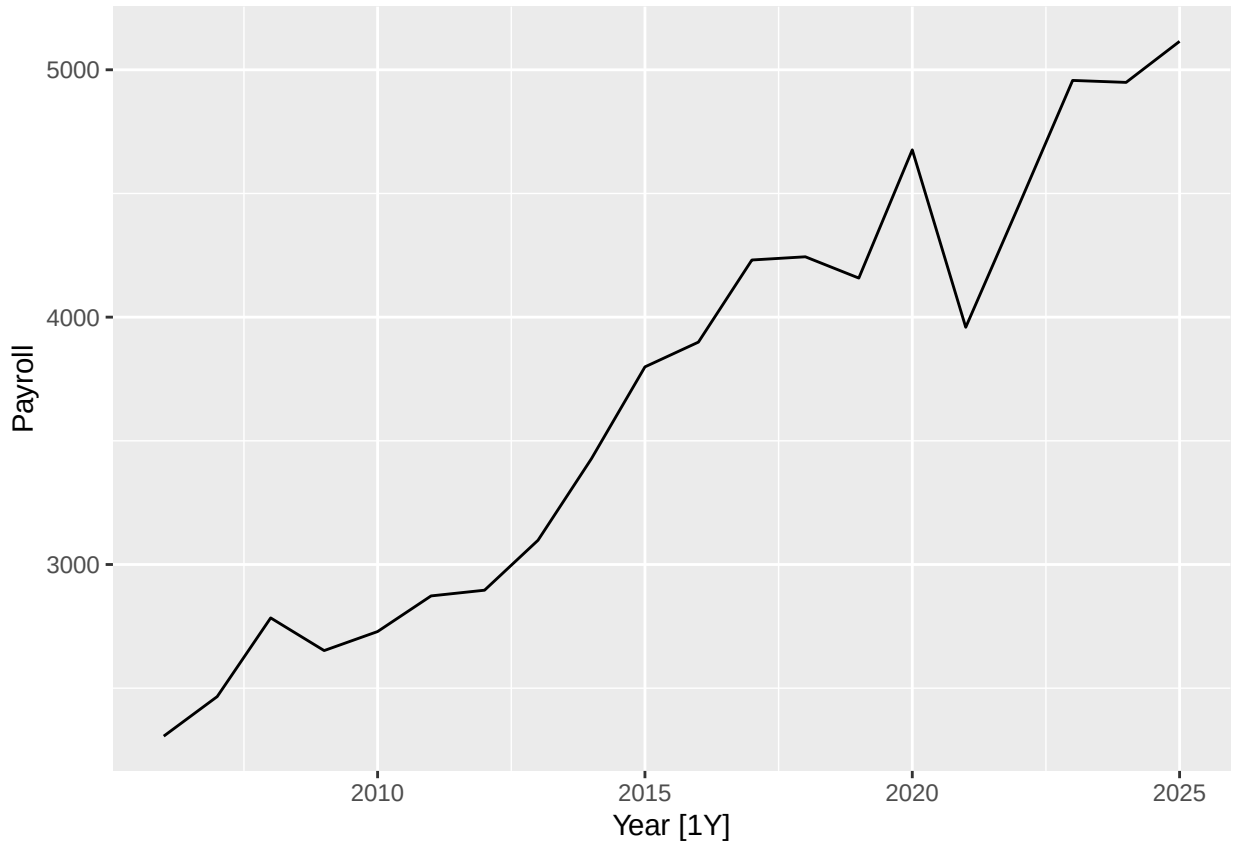
OPS_ts <- OPS_data |>
    pivot_longer(-Year, names_to = "Team", values_to = "OPS") |>
    as_tsibble(key = Team, index = Year)

ERA_ts <- ERA_data |>
    pivot_longer(-Year, names_to = "Team", values_to = "ERA") |>
    as_tsibble(key = Team, index = Year)

autoplot(wins_ts)
```




```
payroll_ts |> filter(Team == "Total") |> autoplot()
```




Here I have gathered the data that I am going to use. Wins in a season, Payroll, OPS, and ERA in a season for each team. I will be focusing my investigation on the total MLB payroll which I have plotted. There is a clear upward trend, with some variation in the data.

**Results**

```
# Get year range from data (e.g. payroll_total or any tsibble with Year)
payroll_total <- payroll_ts |> filter(Team == "Total")
years <- sort(unique(payroll_total$Year))
train_years <- years[1:14]
test_years <- years[15:length(years)]

payroll_train <- payroll_total |> filter(Year %in% train_years)
payroll_test <- payroll_total |> filter(Year %in% test_years)
```

First, I am going to split into the train and test data which I will use to train and test a variety of models.

**ARIMA**

```
# Best ARIMA on train data (auto.arima selects order)
payroll_arima_fit <- auto.arima(payroll_train$Payroll)
summary(payroll_arima_fit)
```

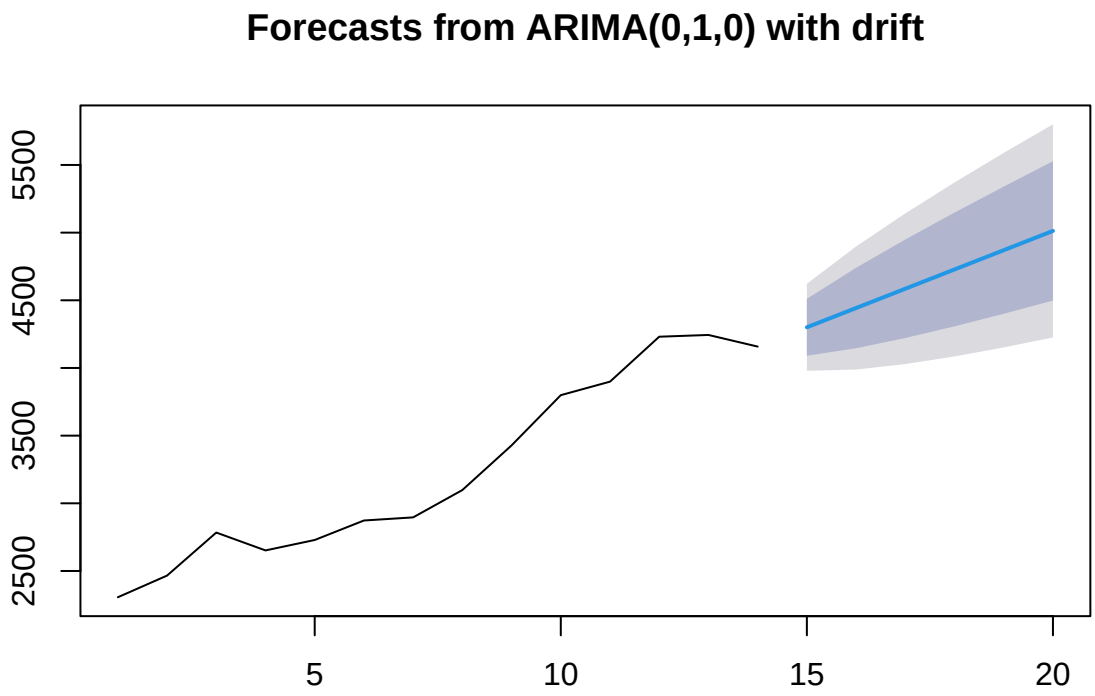
```
## Series: payroll_train$Payroll
## ARIMA(0,1,0) with drift
##
```

```
## Coefficients:
##          drift
##       142.4615
## s.e.   43.6934
##
## sigma^2 = 26887:  log likelihood = -84.22
## AIC=172.44   AICc=173.64   BIC=173.57
##
## Training set error measures:
##                      ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.1545384 151.8086 122.9787 -0.1039322 3.682785 0.6987427
##                    ACF1
## Training set 0.02102863
```

```
# Forecast test horizon and get accuracy vs test data
payroll_arima_fc <- forecast(payroll_arima_fit, h = length(test_years))
# Summary of ARIMA on test data (RMSE, MAE, etc.)
accuracy(payroll_arima_fc, payroll_test$Payroll)
```

```
##                      ME     RMSE      MAE        MPE     MAPE      MASE
## Training set  0.1545384 151.8086 122.9787 -0.1039322 3.682785 0.6987427
## Test set     28.6179487 277.3886 233.3872  0.1834220 5.233445 1.3260635
##                    ACF1
## Training set 0.02102863
## Test set             NA
```

```
plot(payroll_arima_fc, payroll_test$Payroll)
```




I used auto.arima on the total MLB payroll data. It chose an ARIMA(0,1,0) with drift, which is a random walk with drift (the data has been differenced once). The forecast shows an upward trend and the test data has a MAPE of 5.23% which is the mean absolute percent error, or that the error was on average 5% off from the predicted value.

**ETS**

```
payroll_ets_fit <- ets(payroll_train$Payroll)
summary(payroll_ets_fit)
```

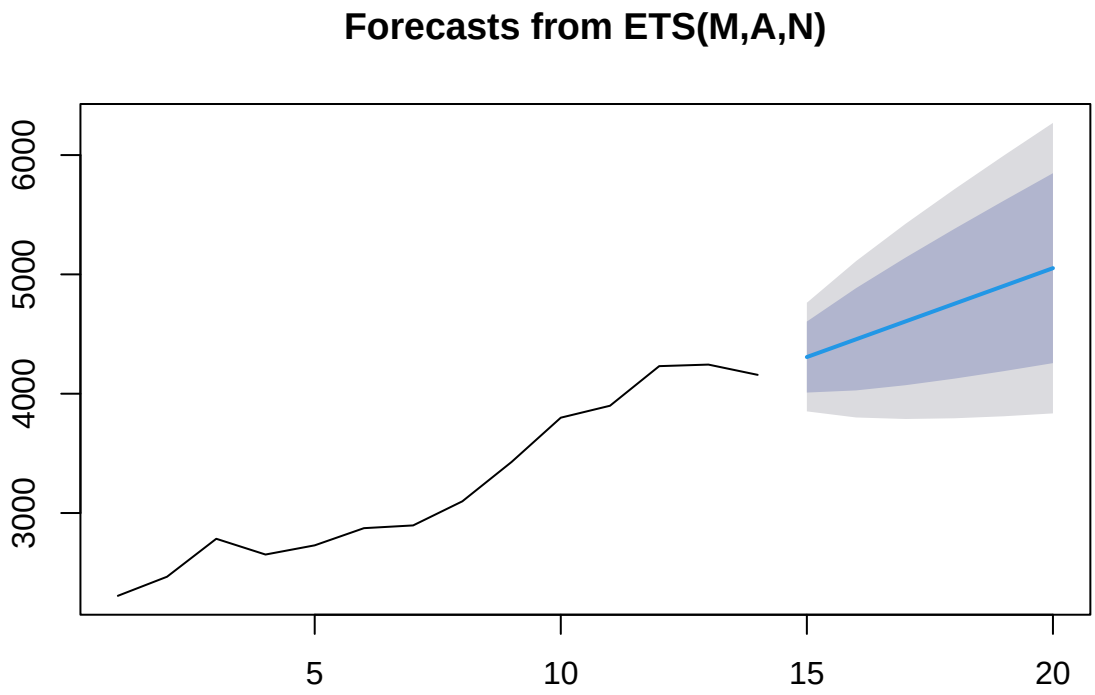
```
## ETS(M,A,N)
##
## Call:
## ets(y = payroll_train$Payroll)
##
##   Smoothing parameters:
##     alpha = 0.9996
##     beta  = 1e-04
##
##   Initial states:
##     l = 2148.8217
##     b = 149.0317
##
##   sigma:  0.0539
##
##      AIC     AICc      BIC
```

```
## 186.3983 193.8983 189.5935
##
## Training set error measures:
##                     ME     RMSE      MAE        MPE     MAPE    MASE       ACF1
## Training set -5.513547 151.9654 123.6629 -0.2752598 3.707549 0.70263 0.02138063
```

```
payroll_ets_fc <- forecast(payroll_ets_fit, h = length(test_years))
accuracy(payroll_ets_fc, payroll_test$Payroll)
```

```
##                     ME     RMSE      MAE        MPE     MAPE    MASE       ACF1
## Training set -5.513547 151.9654 123.6629 -0.2752598 3.707549 0.70263 0.02138063
## Test set      5.559031 274.8447 221.3271 -0.2978770 5.011424 1.25754         NA
```

```
plot(payroll_ets_fc, payroll_test$Payroll)
```




ETS was very similar to the ARIMA in terms of errors, having MAPE on the test data of 5.01%, slightly beating it out. ETS fitted a model with multiplicative errors, additive trend, and no seasonality.

**Holt-Winters**

```
payroll_hw_fit <- hw(payroll_train$Payroll)
```

```
## Error in hw(payroll_train$Payroll): The time series should have frequency greater than 1.
```

```
summary(payroll_hw_fit)
```

```
## Error in h(simpleError(msg, call)): error in evaluating the argument 'object' in selecting a method
```

```
payroll_hw_fc <- forecast(payroll_hw_fit, h = length(test_years))
```

```
## Error: object 'payroll_hw_fit' not found
```

```
accuracy(payroll_hw_fc, payroll_test$Payroll)
```

```
## Error: object 'payroll_hw_fc' not found
```

```
plot(payroll_hw_fc, payroll_test$Payroll)
```

```
## Error in h(simpleError(msg, call)): error in evaluating the argument 'x' in selecting a method for fu
```

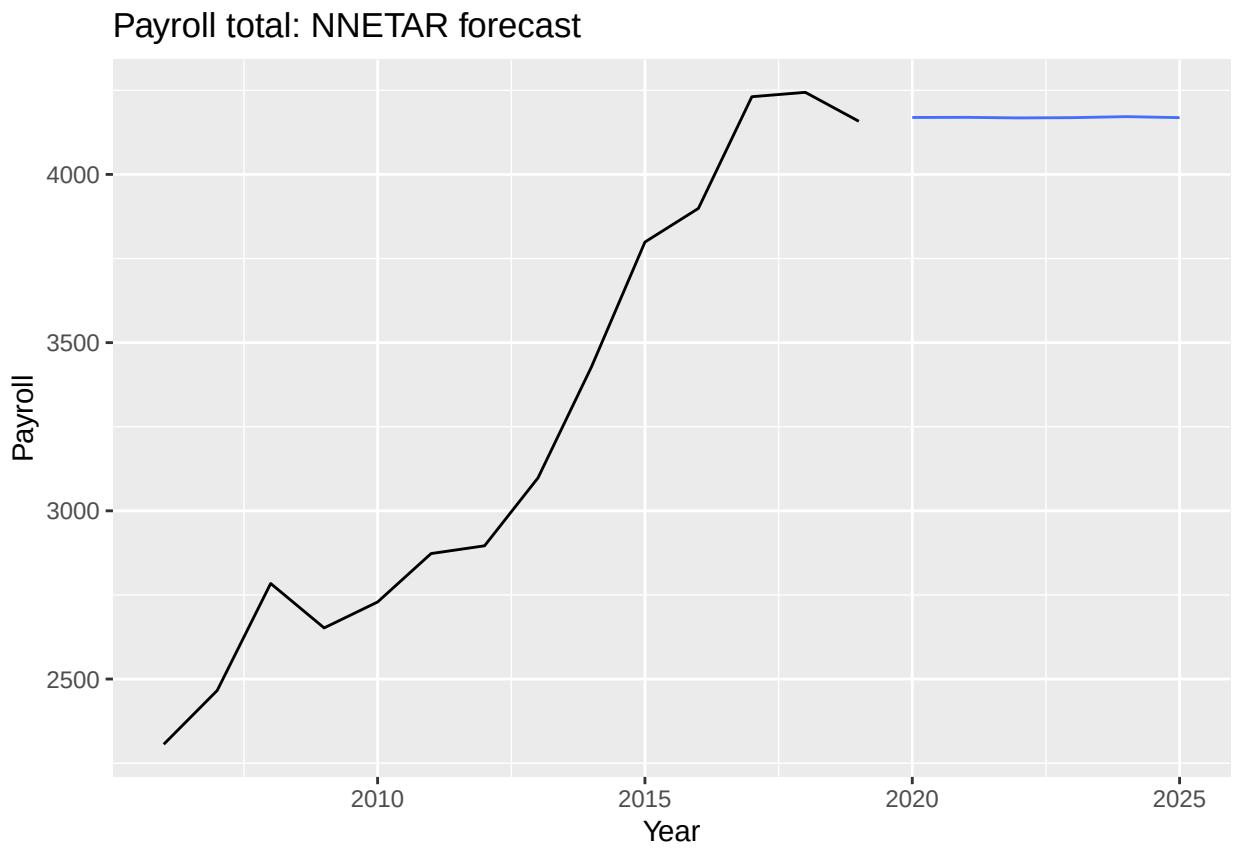
The Holt-Winters cannot be estimated because the data is annual (frequency of 1), so there is no seasonal component in the data.

**NNETAR**

```
# fable: tsibble + model(NNETAR(Payroll)), no transformation
set.seed(144)
payroll_nnetar_fit <- payroll_train |>
  model(NNETAR(Payroll))
payroll_nnetar_fit
```

```
## # A mable: 1 x 2
## # Key:     Team [1]
##   Team  `NNETAR(Payroll)`
##   <chr>           <model>
## 1 Total       <NNAR(1,1)>
```

```
payroll_nnetar_fit |>
  forecast(h = length(test_years)) |>
  autoplot(payroll_train, level = NULL) +
  labs(x = "Year", y = "Payroll", title = "Payroll total: NNETAR forecast")
```




```
payroll_nnetar_fc <- payroll_nnetar_fit |> forecast(h = length(test_years))
accuracy(payroll_nnetar_fc, payroll_test)
```

```
## # A tibble: 1 x 11
##   .model          Team  .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
##   <chr>           <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 NNETAR(Payroll) Total Test   515.  646.  585.  10.3  12.1   NaN   NaN 0.329
```

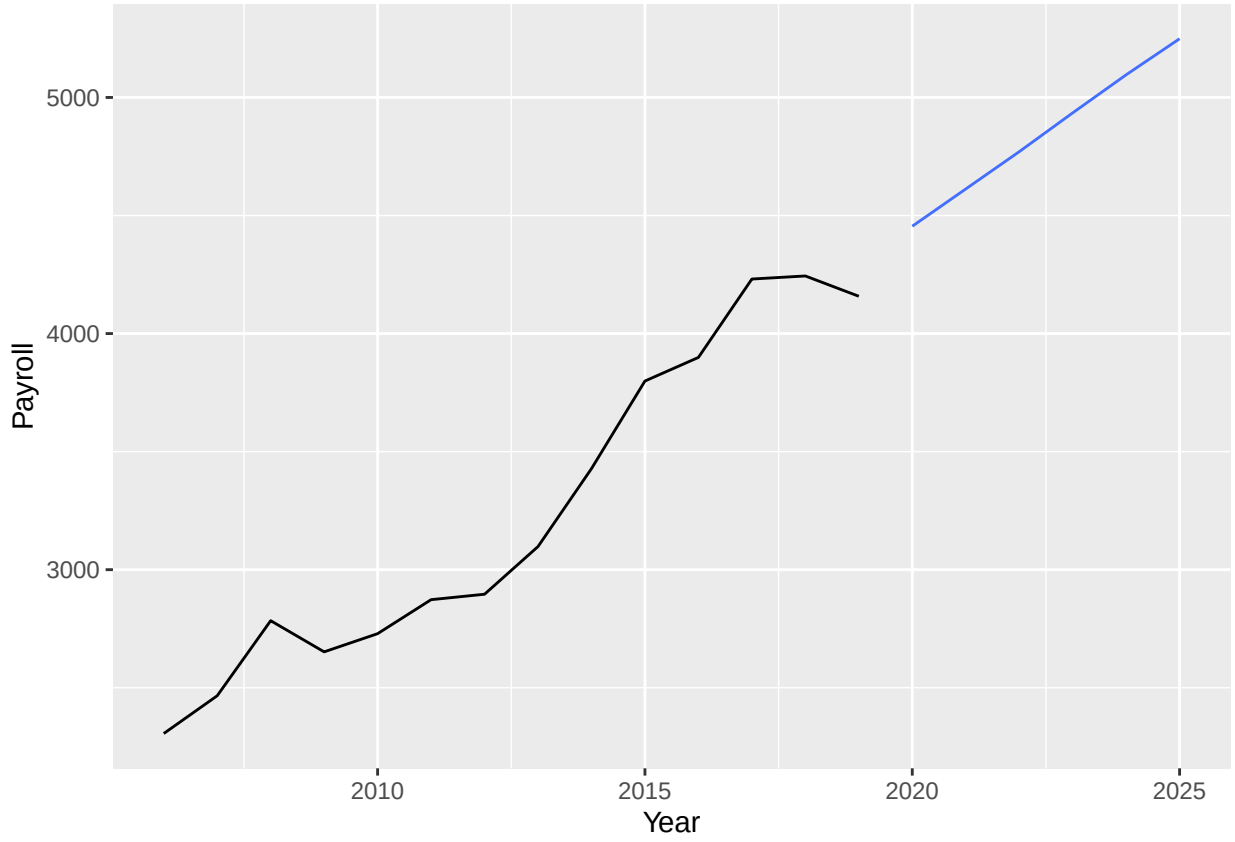
NNETAR fitted a NNAR(1,1) which has 1 lag and 1 node. NNETAR did not perform well in terms of MAPE, where the test MAPE was 12.1%, far worse than other models seen so far, for this reason I will likely not use it in my combined forecast.

**Prophet**

```
# fable.prophet: fit Prophet inside model() like ARIMA/ETS (annual data: no season term)
payroll_prophet_fit <- payroll_train |>
  model(prophet = prophet(Payroll))
payroll_prophet_fit
```

```
## # A mable: 1 x 2
## # Key:     Team [1]
##   Team    prophet
##   <chr>   <model>
## 1 Total <prophet>
```

```
# Forecast and plot (same pattern as other models)
payroll_prophet_fc <- payroll_prophet_fit |> forecast(h = length(test_years))
payroll_prophet_fc |> autoplot(payroll_train, level = NULL)
```




```
# Accuracy vs test
accuracy(payroll_prophet_fc, payroll_test)
```

```
## # A tibble: 1 x 11
##   .model  Team  .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE   ACF1
##   <chr>   <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>
## 1 prophet Total Test  -167.  320.  249. -3.99  5.73   NaN   NaN -0.315
```

Prophet performed well with a MAPE of 5.7% and looked to be very similar to both ARIMA and ETS in terms of a forecast.

**All models: one-year-ahead forecast (2026)**

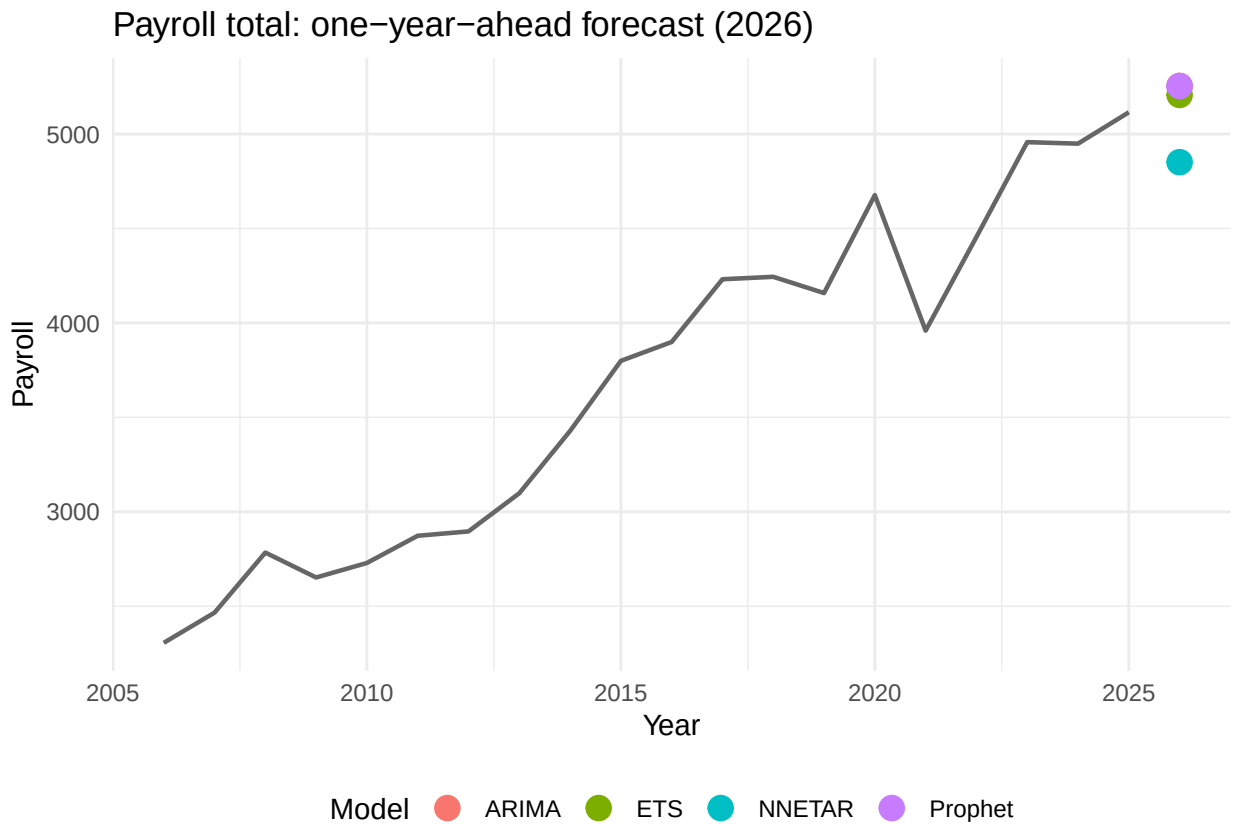
Here is a plot of how all of the models would predict a 2026 payroll data, trained on the entire dataset from 2006-2025.

```
fc_year <- max(payroll_total$Year) + 1
set.seed(144)

arima_fc1   <- forecast(auto.arima(payroll_total$Payroll), h = 1)$mean[1]
ets_fc1     <- forecast(ets(payroll_total$Payroll), h = 1)$mean[1]
```

```
nnetar_fc1  <- (payroll_total |> model(NNETAR(Payroll)) |> forecast(h = 1))$.mean[1]
prophet_fc1 <- (payroll_total |> model(prophet = prophet(Payroll)) |> forecast(h = 1))$.mean[1]

fc_tbl <- tibble(
  Model = c("ARIMA", "ETS", "NNETAR", "Prophet"),
  Year = fc_year,
  Payroll = c(arima_fc1, ets_fc1, nnetar_fc1, prophet_fc1)
)
ggplot() +
  geom_line(data = payroll_total, aes(x = Year, y = Payroll), linewidth = 0.8, colour = "gray40") +
  geom_point(data = fc_tbl, aes(x = Year, y = Payroll, colour = Model), size = 4) +
  labs(x = "Year", y = "Payroll", title = "Payroll total: one-year-ahead forecast (2026)") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

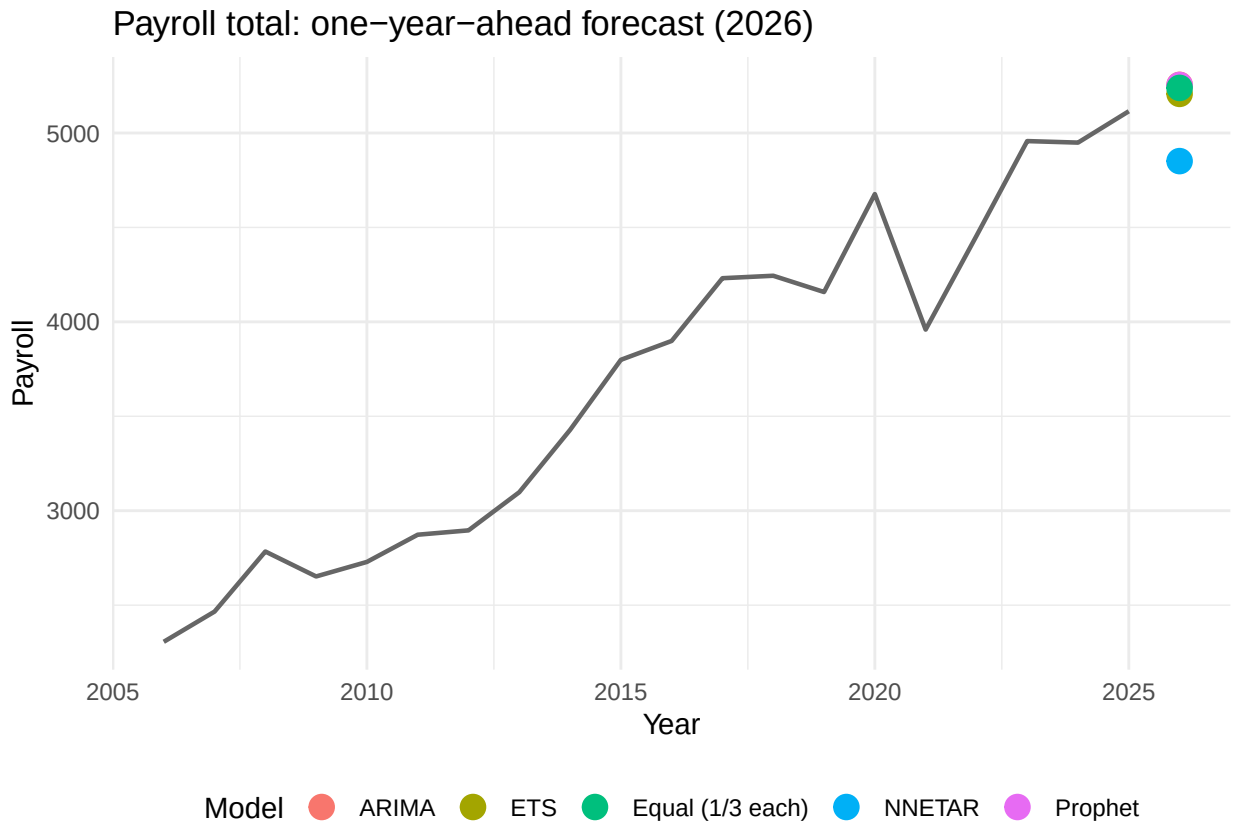



**Forecast Combination**

For a combined forecast, since Prophet, ARIMA, and ETS all performed similarly and well, I will be using a combination of the three of them to forecast one year ahead for 2026 payrolls. For weights, I will use equal weights between them.

```
w_equal <- c(ARIMA = 1/3, ETS = 1/3, Prophet = 1/3)
# 2026 forecast from equal-weight combination
equal_weight_fc_2026 <- w_equal["ARIMA"] * arima_fc1 + w_equal["ETS"] * ets_fc1 + w_equal["Prophet"] *
# Add to plot
fc_tbl <- bind_rows(
```

```
  fc_tbl,
  tibble(Model = "Equal (1/3 each)", Year = fc_year, Payroll = as.numeric(equal_weight_fc_2026))
)
ggplot() +
  geom_line(data = payroll_total, aes(x = Year, y = Payroll), linewidth = 0.8, colour = "gray40") +
  geom_point(data = fc_tbl, aes(x = Year, y = Payroll, colour = Model), size = 4) +
  labs(x = "Year", y = "Payroll", title = "Payroll total: one-year-ahead forecast (2026)") +
  theme_minimal() +
  theme(legend.position = "bottom")
```




**Other**

Now, I will explore the relationship between wins and payroll, looking deeper at my favorite team, the Padres.

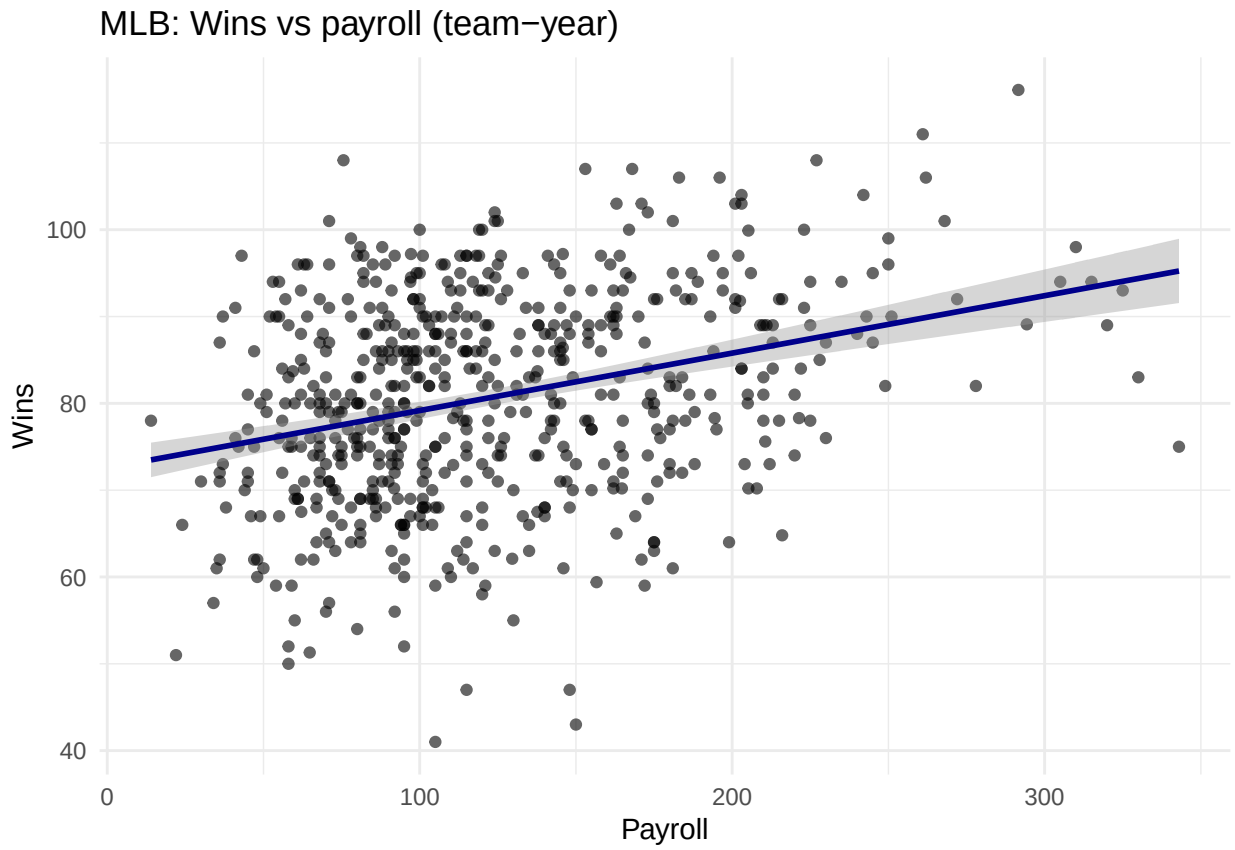
**Wins vs payroll (entire MLB, team-year)**

```
# Join wins and payroll by Team and Year; one row per team per year
mlb_wins_payroll <- wins_ts |>
  inner_join(payroll_ts |> dplyr::select(Year, Team, Payroll), by = c("Year", "Team"))
# Simple regression: Wins on Payroll
fit_wins_payroll <- lm(Wins ~ Payroll, data = mlb_wins_payroll)
summary(fit_wins_payroll)
```

```
##
## Call:
```

```
## lm(formula = Wins ~ Payroll, data = mlb_wins_payroll)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -39.479  -8.154   0.046   8.210  30.442
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 72.558641   1.117632  64.922  < 2e-16 ***
## Payroll      0.066134   0.008264   8.003 6.35e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.47 on 598 degrees of freedom
## Multiple R-squared:  0.09674,    Adjusted R-squared:  0.09523
## F-statistic: 64.04 on 1 and 598 DF,  p-value: 6.347e-15
```

```r
# Scatter plot: Wins vs Payroll with fitted line
ggplot(mlb_wins_payroll, aes(x = Payroll, y = Wins)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, colour = "darkblue") +
  labs(x = "Payroll", y = "Wins", title = "MLB: Wins vs payroll (team-year)") +
  theme_minimal()
```




Based on the simple regression, increasing payroll by $1 million increased expected wins by .066 games, and the very small p-value shows this effect is statistically significant. This means that teams like the Dodgers

who pay their players $150 million more per year compared to the median team, are expected to have 10 more regular season wins compared to the median team, a significant amount.

```
# Padres data: join Wins, OPS, ERA, Payroll by Year
padres_df <- wins_ts |>
  filter(Team == "Padres") |>
  dplyr::select(Year, Wins) |>
  left_join(OPS_ts |> filter(Team == "Padres") |> dplyr::select(Year, OPS), by = "Year") |>
  left_join(ERA_ts |> filter(Team == "Padres") |> dplyr::select(Year, ERA), by = "Year") |>
  left_join(payroll_ts |> filter(Team == "Padres") |> dplyr::select(Year, Payroll), by = "Year") |>
  arrange(Year) |>
  na.omit()

# ARDL: Wins on own lags and lags of OPS, ERA, Payroll (dynlm)
padres_ts <- zoo::zoo(padres_df[, c("Wins", "OPS", "ERA", "Payroll")], order.by = padres_df$Year)

ardl_padres <- dynlm(Wins ~ L(Wins, 1:2) + L(OPS, 1:2) + L(ERA, 1:2) + L(Payroll, 1:2), data = padres_ts
summary(ardl_padres)
```

```
##
## Time series regression with "zoo" data:
## Start = 2013, End = 2025
##
## Call:
## dynlm(formula = Wins ~ L(Wins, 1:2) + L(OPS, 1:2) + L(ERA, 1:2) +
##     L(Payroll, 1:2), data = padres_ts)
##
## Residuals:
##   [1]   9.4190   0.5163  -1.7150 -11.7582  -3.4904 -13.1903   1.6759  15.1956
##   [9]  -0.7459   3.5534  -1.4145   1.4006   0.5534
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         28.1754   152.1178   0.185    0.862
## L(Wins, 1:2)1        1.0615     1.7334   0.612    0.573
## L(Wins, 1:2)2        1.5422     1.7348   0.889    0.424
## L(OPS, 1:2)1       -42.0273   276.3355  -0.152    0.886
## L(OPS, 1:2)2      -356.0564   348.7291  -1.021    0.365
## L(ERA, 1:2)1        21.8680    43.7315   0.500    0.643
## L(ERA, 1:2)2        11.1642    22.9252   0.487    0.652
## L(Payroll, 1:2)1    -0.0728     0.1933  -0.377    0.726
## L(Payroll, 1:2)2     0.0673     0.1582   0.426    0.692
##
## Residual standard error: 12.92 on 4 degrees of freedom
## Multiple R-squared:  0.4985, Adjusted R-squared:  -0.5045
## F-statistic: 0.497 on 8 and 4 DF,  p-value: 0.8144
```

After running an ARDL on Padres wins with past lags of Wins, ERA, OPS, and Payroll, nothing came back as significant.

```
padres_payroll=dynlm(Wins ~ L(Payroll, 0), data=padres_ts)
summary(padres_payroll)
```

```
##
## Time series regression with "zoo" data:
## Start = 2011, End = 2025
##
## Call:
## dynlm(formula = Wins ~ L(Payroll, 0), data = padres_ts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3665  -5.4796   0.9322   4.8439  12.4883
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   64.19907    4.44580  14.440  2.2e-09 ***
## L(Payroll, 0)  0.11312    0.03118   3.629  0.00306 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.336 on 13 degrees of freedom
## Multiple R-squared:  0.5032, Adjusted R-squared:  0.465
## F-statistic: 13.17 on 1 and 13 DF,  p-value: 0.00306
```

```
padres_lag_payroll=dynlm(Wins ~ L(Payroll, 0:1), data=padres_ts)
summary(padres_lag_payroll)
```

```
##
## Time series regression with "zoo" data:
## Start = 2012, End = 2025
##
## Call:
## dynlm(formula = Wins ~ L(Payroll, 0:1), data = padres_ts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6585  -5.4790   0.0008   5.4700  11.8598
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)      63.756297   5.424949  11.752 1.44e-07 ***
## L(Payroll, 0:1)0  0.121721   0.056169   2.167    0.053 .
## L(Payroll, 0:1)1 -0.006665   0.053062  -0.126    0.902
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.949 on 11 degrees of freedom
## Multiple R-squared:  0.4825, Adjusted R-squared:  0.3885
## F-statistic: 5.129 on 2 and 11 DF,  p-value: 0.02669
```

```
padres_OPS=dynlm(Wins ~ L(OPS, 1), data=padres_ts)
summary(padres_OPS)
```

```
##
## Time series regression with "zoo" data:
```

```
## Start = 2012, End = 2025
##
## Call:
## dynlm(formula = Wins ~ L(OPS, 1), data = padres_ts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.5402  -7.0609  -0.0241   6.2244  19.4747
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.431     56.151  -0.043    0.966
## L(OPS, 1)    114.918     78.818   1.458    0.171
##
## Residual standard error: 9.751 on 12 degrees of freedom
## Multiple R-squared:  0.1505, Adjusted R-squared:  0.0797
## F-statistic: 2.126 on 1 and 12 DF,  p-value: 0.1705
```

```
padres_ERA=dynlm(Wins ~ L(ERA, 1), data=padres_ts)
summary(padres_ERA)
```

```
##
## Time series regression with "zoo" data:
## Start = 2012, End = 2025
##
## Call:
## dynlm(formula = Wins ~ L(ERA, 1), data = padres_ts)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -12.600  -7.364  -2.987   7.924  21.221
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   83.880     30.899   2.715   0.0188 *
## L(ERA, 1)     -1.131      7.680  -0.147   0.8854
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.57 on 12 degrees of freedom
## Multiple R-squared:  0.001803,   Adjusted R-squared:  -0.08138
## F-statistic: 0.02167 on 1 and 12 DF,  p-value: 0.8854
```

```
auto.arima(padres_df$Wins)
```

```
## Series: padres_df$Wins
## ARIMA(0,1,0)
##
## sigma^2 = 123.6:  log likelihood = -53.59
## AIC=109.17   AICc=109.51   BIC=109.81
```

After regressing Padres wins on current and one lag of payroll, I found that Padres wins are correlated with current payroll (significant on its own, and borderline with a p-value of .053 once last year's payroll is

added), but not past payroll. One lag of ERA or OPS cannot be statistically significantly used to predict future Wins.

**Conclusions**

MLB Payrolls are increasing, and this increase leads teams to have a significant advantage over other teams who do not pay. Out of all models, NNETAR was the one with the highest test errors and was much worse compared to ARIMA, ETS, and Prophet who had very similar forecasts and test errors.

There was not much correlation between the Padres performance one year and metrics of their performance in years prior, showing that team changes year by year and does not stay consistent, something that I agree with as a fan of the team.

In terms of future work, I would be interested to see a cointegrated relationship between divisional teams, teams that play each other often and should have a steadier total wins and losses and see how this could affect wins.